



Topological-aware multiple kernel learning for ensemble clustering

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ABSTRACT

Ensemble clustering integrates a set of base clustering results to obtain more accurate consensus results. Most existing methods transform base clustering results into a co-association (CA) matrix, which describes the frequency with which samples belong to the same cluster. However, these methods still exhibit insufficient learning of sample similarity relationships, overlooking the capture of a more comprehensive similarity relationship from multiple perspectives. To address this issue, we propose a method called **t**Opological-aware **M**ultiple **k**Ernel Learning for Ensemble clusTering (OMELET). Specifically, OMELET uses multiple kernel functions to remeasure the similarity between samples from multiple perspectives. To fully exploit the sample similarity relationships in different kernel spaces, OMELET employs multiple kernel learning to learn a consensus kernel and obtain the similarity matrix between samples. To eliminate noise from the similarity matrix captured in the fused kernel space, OMELET leverages topological similarity propagated among highly similar neighbors to reshape the similarity graph. Finally, OMELET captures a topological affinity matrix that represents the sample similarity relationships. Moreover, to enable OMELET to scale on large datasets, we derive a variant called OMELET-SV, which leverages anchors to construct an anchor-based kernel matrix, thereby improving computational efficiency. Extensive experimental results demonstrate the outstanding performance of OMELET compared to 18 representative methods.

1. Introduction

Ensemble clustering aims to integrate a set of base clustering results to obtain a more accurate and robust result. Ensemble clustering is applied in various fields, including medical diagnosis [1], community detection [2], and oil price prediction [3]. The base clustering results can be generated by different clustering algorithms or by the same clustering algorithm with different hyper-parameters [4]. Moreover, the original features are transparent to ensemble clustering [5], which endows ensemble clustering with feature-free characteristics, making it more challenging compared to other clustering algorithms [6].

Due to the limitations of base clustering algorithms, the generated base clustering results may contain a large amount of incorrect labels. Therefore, directly aligning base clustering results containing a large number of incorrect labels may lead to significant bias. Most existing ensemble methods transform base clustering results into the co-association (CA) matrix, which reflects the frequency with which each pair of samples is grouped into the same cluster by the base clustering algorithms. These methods learn sample similarity in two ways. The first method treats the frequency of samples being in the same cluster as

their similarity and captures a more refined and reliable similarity relationship through graph learning or other methods. The second method remeasures the similarity relationships between samples using filtering or extraction techniques. Regardless, these methods still fall short in fully considering sample similarity, lacking the ability to capture similarity from multiple perspectives. We use [Example 1](#) to illustrate this problem.

Example 1. [Fig. 1](#) provides a comparison between single-view and multi-view similarity learning based on the CA matrix. Existing single-view methods measure similarity solely based on the inherent properties of the CA matrix (i.e., the frequency with which samples are co-assigned to the same cluster) and accordingly learn a weighted or redefined CA matrix. However, such methods still struggle to capture information that is absent from the CA matrix (e.g., the similarity between sample 1 and sample 3). In contrast, by leveraging kernel techniques to re-evaluate sample similarities in multiple kernel spaces, the similarity information is effectively enriched. The resulting multiple similarity matrices capture more diverse relationships among samples, which helps recover a more complete structure of sample similarities.

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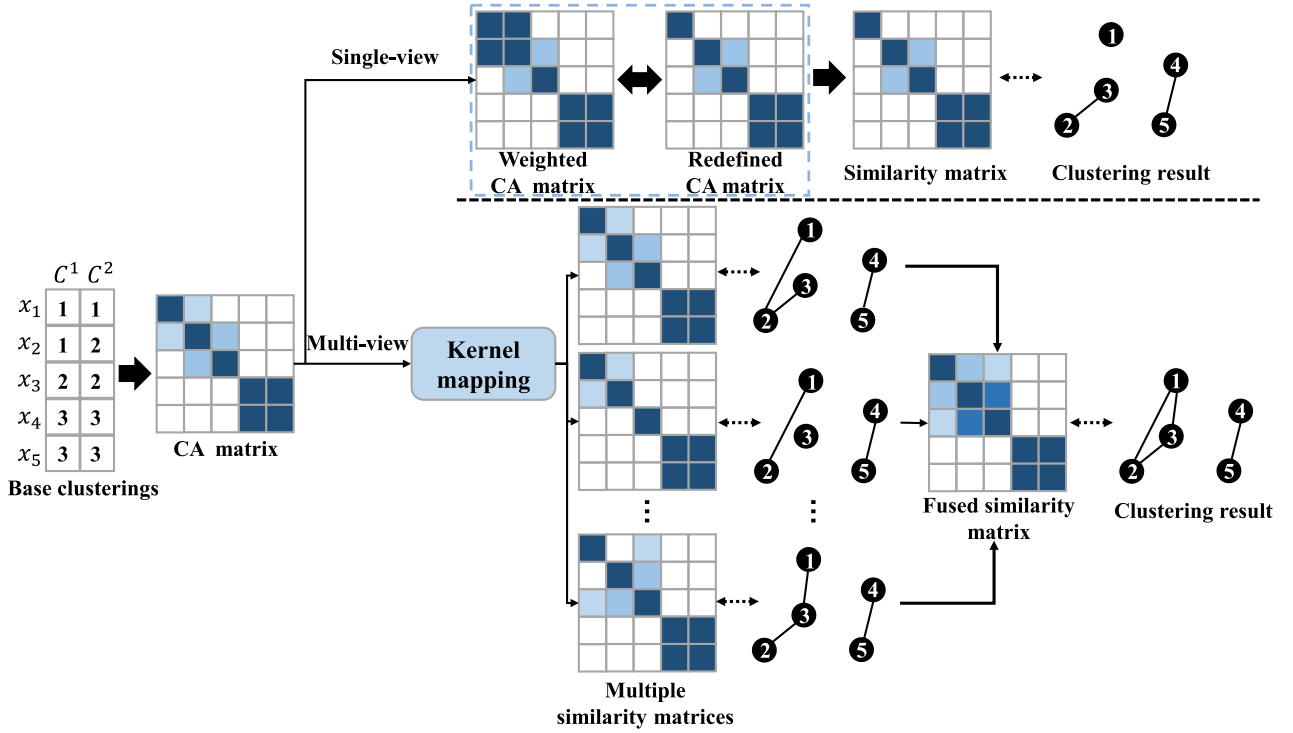


Fig. 1. Comparison between single-view and multi-view of sample similarity.

To address this issue, we propose a method called topological-aware multiple kernel learning for ensemble clustering (OMELET). The motivation for the OMELET method is: (i) The kernel learning strategy can map the rows of the CA matrix as sample features to a Reproducing Kernel Hilbert Space (RKHS), where the similarity relationships between samples may change (not relying on the raw sample features helps maintain the feature-free capability). Multiple kernel learning enables OMELET to capture different sample similarity relationships from multiple perspectives and learn a consensus kernel to capture more stable sample similarity relationships. In other words, if the similarity relationships between samples remain consistent in the multiple kernel space, the likelihood that they belong to the same class is higher. (ii) The sample similarity relationships learned from the consensus kernel (fusion of kernel spaces) may contain noise. Topological similarity learning aims to propagate the topological similarity of samples through highly similar neighbors and can remove noise. When samples i and j are highly similar, applying topological similarity learning helps align the relationships of all other samples with samples i and j . Specifically, when sample k has a noisy connection with sample j , topological consistency will remove this noisy connection, and sample k will have no connection with both i and j .

OMELET first forms a set of base kernels based on the CA matrix. To capture the sample similarity relationships contained in each kernel, OMELET learns a consensus kernel by fusing base kernels and applies an adaptive weighting method to weight the base kernels. Then, OMELET captures the sample similarity matrix by performing self-expression learning within the consensus kernel. Next, OMELET will eliminate noise by capturing the topological similarity between samples. Through topological similarity learning, OMELET can capture a more compact, denoised topological affinity matrix. Finally, spectral clustering is applied to the topological affinity matrix to obtain the consensus result. Compared to other ensemble clustering methods, OMELET is unique in that it integrates similarity relationships from multiple kernel perspectives and enhances noise robustness by capturing topological similarity between samples, which is crucial for improving ensemble clustering accuracy. Note that OMELET relies only on base clustering results with-

out accessing original sample features, thus preserving the feature-free nature of ensemble clustering. Furthermore, to enhance the scalability of the OMELET method for large-scale datasets, we propose OMELET-SV. It employs anchors to construct a set of anchor-based kernel matrices and captures the topological affinity matrix among anchors. Then, the anchor topological affinity matrix is used to construct the sample-anchor similarity matrix, upon which the K-means algorithm is applied to obtain the clustering results, thereby improving computational efficiency.

The main contributions of this paper are listed as follows:

- We propose a novel ensemble clustering method, OMELET, which integrates multiple kernel learning and topological similarity learning into a unified framework.
- OMELET captures the sample similarity relationships from multiple perspectives and designs adaptive base kernel weights to learn a consensus kernel, alleviating the problem of selecting an appropriate kernel function.
- OMELET utilizes topological similarity learning to eliminate noise from the similarity matrix captured in the fused kernel space. A comparison of experimental results with 18 representative methods on 13 datasets demonstrates the effectiveness of OMELET.
- A scalable version OMELET-SV is proposed for large-scale datasets. It leverages anchors to construct the anchor-based kernel matrix, thereby improving computational efficiency.

2. Related work

2.1. Ensemble clustering

In general, ensemble clustering methods can be roughly divided into methods based on base clustering selection and methods based on consensus functions [7]. Many consensus function-based methods enhance ensemble performance by weighting base clustering or introducing learning paradigms and theories [8,9]. For example, Huang *et al.* proposed the local entropy weighting method [10]. Zhou *et al.* introduced the self-paced learning framework into ensemble clustering

[11,12]. Jia *et al.* proposed an ensemble method based on low-rank tensor learning [13] and self-enhancement of CA matrix [14]. Chen *et al.* proposed an ensemble clustering method based on contrastive learning [15]. Iam-On *et al.* proposed a novel link-based ensemble method [16]. Li *et al.* improved ensemble performance by measuring clustering stability [17] and discovering prototype samples [18]. Xu *et al.* introduced high-order connectivity [19] into ensemble clustering. Strehl *et al.* and Zhou *et al.* utilized hypergraph learning to partition consensus results [20]. Chen *et al.* and Huang *et al.* proposed an ensemble clustering method utilizing graph tensor learning [21] and factor graph [22]. Additionally, there are also some ensemble methods based on information theory [23].

In addition to the aforementioned consensus function-based methods, some researchers are also dedicated to selecting appropriate base clusterings for ensemble. A key support for these methods is that the performance of ensemble clustering depends on the diversity and accuracy of the base clustering [17,24]. For example, Huang *et al.* proposed an ensemble hierarchical clustering algorithm based on clustering consistency selection [25]. Yu *et al.* treated the base clustering results as features and employed feature selection techniques to perform the selection of clustering solutions [26]. Abbasi *et al.* proposed a metric called ENMI for ensemble selection [27]. Zhao *et al.* used five internal validity metrics to select base clustering results [28].

2.2. Multiple kernel clustering

Kernel learning is a widely recognized technique for handling nonlinear data, yet selecting high-quality kernels poses a challenge. Multiple kernel learning addresses this issue by automatically extracting discriminative information from a set of kernels, thus circumventing the problem of kernel selection. Multiple Kernel Clustering (MKC) aims to partition data into different categories by optimizing the integration of information from a set of base kernels [29]. Existing MKC methods can be roughly categorized into three types. (i) The first type of method involves integrating low-rank optimization to construct a consensus kernel. For example, Du *et al.* learned a consensus low-rank kernel by removing the unique noise structures from different kernels [30]. (ii) The second type of method utilizes generating high-quality kernel graphs from each individual kernel to compute their clustering results. For example, Ren *et al.* learned multiple candidate affinity graphs from the kernel pool and learned a consistent affinity graph through automatic weighted fusion [31]. Wang *et al.* proposed a multiple kernel K-means method that utilizes original kernel information to guide the fusion of all base partitions [32]. (iii) The third type of method learns a consensus kernel during the clustering process. The basic assumption of this type of method is that the optimal kernel can be represented as a weighted combination of base kernels. For example, Liu *et al.* extended widely used supervised kernel alignment criteria to multiple kernel clustering, and proposed the SimpleMKKM method [33]. Liang *et al.* enabled SimpleMKKM to handle large-scale data by non-asymptotic analysis of kernel weight consistency [34,35]. Beyond kernel fusion, Liu *et al.* proposed contrastive multi-view kernel learning, a kernel generation paradigm for multi-view data that leverages contrastive learning to exploit cross-view complementarity when constructing kernels, and further instantiated it with MKKM for multi-view clustering [36]. Unlike some multiple kernel clustering methods [37] and topological-aware similarity learning methods [38] that are specifically designed for multi-view data, OMELET differs from them in three key aspects: (i) feature-free construction of multiple kernel matrices (without accessing the original input features); (ii) a noise-aware, ensemble-specific topological refinement tailored to mitigating noise in the consensus kernel; and (iii) a scalable variant that makes large-scale clustering feasible in practice.

Table 1

Notations and descriptions used in our method.

Notations	Descriptions
n	Number of samples.
m	Number of base clustering results.
c	Number of clusters.
$\alpha^{(i)}$	The adaptive weight parameters of the i -th base kernel.
$A \in \mathbb{R}^{n \times n}$	The co-association (CA) matrix.
$\tilde{K} \in \mathbb{R}^{n \times n}$	The consensus kernel.
$K^{(i)} \in \mathbb{R}^{n \times n}$	The i -th base kernel.
$Z \in \mathbb{R}^{n \times n}$	The similarity matrix of the samples.
$S \in \mathbb{R}^{n \times n}$	The topological affinity matrix of the samples.
$Y \in \mathbb{R}^{n \times c}$	The cluster indicator matrix.
$L_S \in \mathbb{R}^{n \times n}$	The laplacian matrix of S .

3. The proposed method

3.1. Preliminary

Given n samples $\mathcal{X} = \{x_1, x_2, \dots, x_n\}$ and m base clustering results (i.e. ensemble size) $\Xi = \{C^1, C^2, \dots, C^m\}$, where x_i denotes the i -th sample. $C^j = \{\pi_1^j, \pi_2^j, \dots, \pi_s^j\}$ is the j -th base clustering result, π_h^j is the h -th cluster in C^j , and s is the number of clusters in C^j . The goal of ensemble clustering is to learn m base clustering results to obtain a consensus result C . The mainstream of existing ensemble clustering methods is to transform m base clustering results into a co-association (CA) matrix [39]. The CA matrix describes the frequency with which pairs of samples are assigned to the same cluster and is defined as follows:

$$A_{ij} = \frac{1}{m} \sum_{b=1}^m \psi(\pi^b(x_i), \pi^b(x_j)),$$

$$\psi(\pi^b(x_i), \pi^b(x_j)) = \begin{cases} 1, & \pi^b(x_i) = \pi^b(x_j) \\ 0, & \pi^b(x_i) \neq \pi^b(x_j) \end{cases} \quad (1)$$

where $\pi^b(x_i)$ denotes the cluster of sample x_i in the b -th base clustering. m represents the number of base clustering results, and the range of A_{ij} is $[0, 1]$. The main symbols used in this paper are summarized in Table 1.

3.2. Objective function

To capture diverse sample similarities from multiple perspectives, we construct a set of base kernels $\{K^{(1)}, K^{(2)}, \dots, K^{(k)}\}$ using the CA matrix, where k is the number of base kernels, and $K^{(i)} \in \mathbb{R}^{n \times n}$ represents the i -th base kernel. We utilize the CA matrix to construct base kernels without relying on sample features, preserving the feature-free characteristic of ensemble clustering. The base kernels $K^{(i)}$ are formulated as:

$$K_{jl}^{(i)} = \Phi(A_j)^T \Phi(A_l) = \mathbb{K}(A_j, A_l), \quad (2)$$

where A_j represent the j -th rows of the CA matrix. $\Phi(\cdot) : \mathbb{R}^n \rightarrow \mathbb{H}$ is a mapping function that maps the sample from the original space to the reproducing kernel Hilbert space $\mathbb{H}$, and $\mathbb{K}(\cdot, \cdot)$ is a kernel function. We construct different types of base kernels by applying different kernel functions. Section 4.1 provides the types of kernel functions used.

A straightforward way to learn a consensus kernel is to minimize its discrepancy from the base kernels: $\min_{\tilde{K}} \sum_{i=1}^k \|\tilde{K} - K^{(i)}\|_F^2$, where $\tilde{K}$ denotes consensus kernel. However, this formulation is suboptimal because it treats all base kernels equally and overlooks the differences among them. To highlight each kernel's unique contribution, we introduce adaptive weights and obtain the weighted formulation:

$$\min_{\tilde{K}, \alpha^{(i)}} \sum_{i=1}^k \|\tilde{K} - \alpha^{(i)} K^{(i)}\|_F^2 \text{ s.t. } \sum_{i=1}^k \alpha^{(i)} = 1, \alpha^{(i)} \geq 0, \quad (3)$$

where $\alpha^{(i)}$ is the adaptive weight for the i -th base kernel, and we aim to learn a consensus kernel. After obtaining the consensus kernel, we need to capture the sample similarities within it. We employ a kernel-based

self-expressive model [40] to derive the similarity matrix Z , which is formulated as:

$$\begin{aligned} \min_{\tilde{K}, Z, \alpha^{(i)}} & \underbrace{tr(\tilde{K} - 2Z^T \tilde{K} + Z^T \tilde{K} Z) + \lambda \|Z\|_F^2}_{\text{similarity matrix learning}} + \underbrace{\sum_{i=1}^k \|\tilde{K} - \alpha^{(i)} K^{(i)}\|_F^2}_{\text{adaptive kernel fusion}} \\ \text{s.t. } & \sum_{i=1}^k \alpha^{(i)} = 1, \alpha^{(i)} \geq 0, Z \geq 0, \text{diag}(Z) = 0. \end{aligned} \quad (4)$$

Eq. (4) consists of two parts: the similarity matrix learning term and the consensus kernel learning term. Although some self-expressive models employ l_1 regularization or nuclear-norm (low-rank) penalties, we believe that Frobenius norm regularization is more suitable for OMELET for the following reasons: (i) l_1 and low-rank penalties typically lead to more expensive optimization procedures. (ii) As a stable and lightweight constraint, the Frobenius norm regularizer avoids overly aggressive reshaping of Z (e.g., forcing some rows to be extremely sparse) before the subsequent topological-based denoising step.

Although we obtain the similarity matrix Z through the consensus kernel, some noise is inevitably introduced during the learning process and propagated to Z . To eliminate the noise, we introduce topological similarity learning. The formulation of topological similarity learning [41] is as follows:

$$\frac{1}{2} \sum_{i,j,k=1}^n G_{jk} (S_{ij} - S_{ik})^2 + \gamma \|S - I\|_F^2, \quad (5)$$

where S denotes topological affinity matrix of the samples, and I is the identity matrix. G represents a predefined sample similarity graph. The term $\|S - I\|_F^2$ is designed to avoid trivial solutions [41]. Finally, our objective function is as follows:

$$\begin{aligned} \min_{\tilde{K}, Z} & \underbrace{tr(\tilde{K} - 2Z^T \tilde{K} + Z^T \tilde{K} Z) + \lambda \|Z\|_F^2}_{\text{similarity matrix learning}} + \underbrace{\sum_{i=1}^k \|\tilde{K} - \alpha^{(i)} K^{(i)}\|_F^2}_{\text{adaptive kernel fusion}} \\ & + \underbrace{\frac{1}{2} \sum_{i,j,k=1}^n Z_{jk} \left(\frac{S_{ij}}{\sqrt{D_{jj}}} - \frac{S_{ik}}{\sqrt{D_{kk}}} \right)^2 + \gamma \|S - I\|_F^2}_{\text{topological similarity learning}} \\ \text{s.t. } & Z \geq 0, \text{diag}(Z) = 0, \text{rank}(L_S) = n - c, \\ & \sum_{i=1}^k \alpha^{(i)} = 1, \alpha^{(i)} \geq 0, S_i^T \mathbf{1} = 1, S_{ij} \geq 0, \end{aligned} \quad (6)$$

where $D_{ii} = \sum_{j=1}^n Z_{ij}$, D is the diagonal degree matrix of Z , and the rank constraint on the laplacian matrix L_S of S ensures that S has c connected components. Additionally, S is required to satisfy the conditions that all its elements are greater than 0 and that each row sums to 1. Notice that Eq. (6) has the following characteristics: • It utilizes multiple base kernels to learn a consensus kernel, revealing the diverse similarity relationships between samples. It also does not require access to the original data features, thus inheriting the feature-free property of ensemble clustering. • It learns a topological affinity matrix S among samples through topological similarity learning. Topological similarity learning helps mitigate noise in the consensus kernel (i.e., the fused kernel space) and thus yields a cleaner affinity structure.

3.3. Optimization

To solve Eq. (6), we employ the alternating optimization strategy [42], i.e., we optimize the target variable alternately while fixing other variables.

(1) Update $\tilde{K}$: To update variable $\tilde{K}$, we fix other variables and eliminate those unrelated to $\tilde{K}$. The formula is rewritten as follows:

$$\min_{\tilde{K}} tr(\tilde{K} - 2Z^T \tilde{K} + Z^T \tilde{K} Z) + \sum_{i=1}^k \|\tilde{K} - \alpha^{(i)} K^{(i)}\|_F^2. \quad (7)$$

Since Eq. (7) is unconstrained problem, we take the derivative of the above function w.r.t. $\tilde{K}$ and set the result to 0, obtaining the following result:

$$\tilde{K} = \frac{2 \sum_{i=1}^k \alpha^{(i)} K^{(i)} + 2Z^T - I - ZZ^T}{2k}. \quad (8)$$

After each update of the consensus kernel $\tilde{K}$, we apply a post-processing step to ensure symmetry and positive semidefiniteness. Specifically, we first symmetrize $\tilde{K}$ by setting $\tilde{K} = (\tilde{K} + \tilde{K}^T)/2$. Then we compute the eigendecomposition of the symmetric matrix, $\tilde{K} = U \text{diag}(\lambda_{\tilde{K}}) U^T$, and obtain nonnegative eigenvalues by setting $\lambda_{\tilde{K}} = \max(\lambda_{\tilde{K}}, 0)$. Then, we reconstruct the consensus kernel as $\tilde{K} = U \text{diag}(\lambda_{\tilde{K}}) U^T$.

(2) Update Z : With the other variables fixed, the subproblem for Z is:

$$\begin{aligned} \min_Z & tr(-2Z^T \tilde{K} + Z^T \tilde{K} Z) + \lambda \|Z\|_F^2 + \frac{1}{2} \sum_{i,j,k=1}^n Z_{jk} \left(\frac{S_{ij}}{\sqrt{D_{jj}}} - \frac{S_{ik}}{\sqrt{D_{kk}}} \right)^2 \\ \text{s.t. } & Z \geq 0, \text{diag}(Z) = 0. \end{aligned} \quad (9)$$

For Eq. (9), we take its derivative and then straightforwardly satisfy the constraints, obtaining the following result:

$$Z = \max(0, (2\tilde{K} + 2\lambda I)^{-1} (2\tilde{K} + D^{-\frac{1}{2}} S S^T D^{-\frac{1}{2}})). \quad (10)$$

To ensure symmetry, we symmetrize the sample similarity matrix Z by $Z = (Z^T + Z)/2$.

(3) Update S : The variable S is updated by solving:

$$\begin{aligned} \min_S & \frac{1}{2} \sum_{i,j,k=1}^n Z_{jk} \left(\frac{S_{ij}}{\sqrt{D_{jj}}} - \frac{S_{ik}}{\sqrt{D_{kk}}} \right)^2 + \gamma \|S - I\|_F^2 \\ \text{s.t. } & S_i^T \mathbf{1} = 1, S_{ij} \geq 0, \text{rank}(L_S) = n - c. \end{aligned} \quad (11)$$

Due to the presence of rank constraints, solving Eq. (11) poses certain challenges. However, according to Ky Fan's theorem [43], we can transform the rank constraint into:

$$\min_Y tr(Y^T L_S Y) \quad \text{s.t. } Y^T Y = I. \quad (12)$$

where $Y \in \mathbb{R}^{n \times c}$ is the cluster indicator matrix, I is the identity matrix. Eq. (12) is a spectral problem, and the solution for Y is obtained by the c eigenvectors of L_S corresponding to the c smallest eigenvalues. For the Laplacian matrix L_S , we define $L_S = D_S - (S^T + S)/2$, where D_S is the degree matrix of $(S^T + S)/2$. Next, Eq. (11) is rewritten as:

$$\begin{aligned} \min_S & \frac{1}{2} \sum_{i,j,k=1}^n Z_{jk} \left(\frac{S_{ij}}{\sqrt{D_{jj}}} - \frac{S_{ik}}{\sqrt{D_{kk}}} \right)^2 + \gamma \|S - I\|_F^2 + 2\beta tr(Y^T L_S Y) \\ \text{s.t. } & S_i^T \mathbf{1} = 1, S_{ij} \geq 0, Y^T Y = I. \end{aligned} \quad (13)$$

Eq. (13) can be rewritten as:

$$\min_{S_{ij} \geq 0, S_i^T \mathbf{1} = 1} S_i^T G S_i - S_i^T b, \quad (14)$$

where $G = (I - D^{-\frac{1}{2}} Z D^{-\frac{1}{2}}) + \gamma I$, $b = 2\gamma I_i - \beta \|Y_i - Y_j\|_2^2$. Eq. (14) is a quadratic convex optimization problem that can be solved using the augmented Lagrangian multiplier method (ALM). We transform it into:

$$\min_{S_{ij} \geq 0, S_i^T \mathbf{1} = 1, p = S_i} S_i^T G p - S_i^T b, \quad (15)$$

where p is auxiliary variable. The Lagrangian function w.r.t. Eq. (15) is:

$$\min_{S_{ij} \geq 0, S_i^T \mathbf{1} = 1, p = S_i} S_i^T G p - S_i^T b + \frac{\eta}{2} \|S_i - p + \frac{1}{\eta} q\|_2^2, \quad (16)$$

where q and η are the parameter and penalty coefficient, respectively. According to [44], the variables in Eq. (16) can be alternately fixed and iterated to solve. The corresponding cost function regarding S_i can be formulated as:

$$\min_{S_{ij} \geq 0, S_i^T \mathbf{1} = 1} \|S_i - p + \frac{1}{\eta} q + \frac{Gp - b}{\eta}\|_2^2. \quad (17)$$

The Eq. (17) has a closed-form solution, which can be solved by [45]. Then, the topological affinity matrix S is reconstructed by $S = (S^T + S)/2$. The solution for other variables can be obtained by setting their corresponding derivatives to 0. Algorithm 1 summarizes the procedure for solving Eq. (15).

Algorithm 1 ALM update S .

Input: Z, Y , parameter $\gamma, \beta = 1.5$.

Initialization: $\rho = 1.5, \eta = 10$, and a nonzero vector q .

Output: The topological affinity matrix S .

```

1: while not converged do
2:   Update  $p$  by  $p = S_i - \frac{1}{\eta}(G^T S_i + q)$ ;
3:   Update  $S_i$  by Eq. (17);
4:   Update  $\eta$  by  $\eta = \rho\eta$ ;
5:   Update  $q$  by  $q = q + \eta(S_i - p)$ ;
6: end while

```

(4) **Update** $\alpha^{(i)}$: Update $\alpha^{(i)}$ by solving the following equation:

$$\min_{\alpha^{(i)}} \sum_{i=1}^k \left\| \tilde{K} - \alpha^{(i)} K^{(i)} \right\|_F^2 \quad \text{s.t.} \quad \sum_{i=1}^k \alpha^{(i)} = 1, \alpha^{(i)} \geq 0. \quad (18)$$

Eq. (18) involves both inequality and equality constraints. Here, we derive the Lagrangian function as follows:

$$\mathcal{L} = \left\| \tilde{K} - \alpha^{(i)} K^{(i)} \right\|_F^2 + \delta \left(\sum_{i=1}^k \alpha^{(i)} - 1 \right). \quad (19)$$

where δ denotes the Lagrange multiplier. We directly take the derivative of Eq. (19) and set it to zero, resulting in:

$$\alpha^{(i)} = \frac{2tr(\tilde{K} K^{(i)}) - \delta}{2tr(K^{(i)} (K^{(i)})^T)}. \quad (20)$$

Algorithm 2 OMELET.

Input: A set of kernel matrices $\{K_p^{(1)}, K_p^{(2)}, \dots, K_p^{(k)}\}, \lambda, \gamma$.

Initialization: $S = Z = I, \alpha^{(i)} = 1/k$, and $\epsilon = 10^{-2}$.

Output: Clustering result C .

```

1: while not converged do
2:   Update  $\tilde{K}$  by Eq. (8); Update  $Z$  by Eq. (10);
3:   Update  $Y$  by Eq. (12); Update  $\alpha^{(i)}$  by Eq. (20);
4:   Update  $S$  by Algorithm 1;
5:   Check the convergence conditions:  $(obj(t-1) - obj(t))/obj(t) \leq \epsilon$ ;
6: end while
7: Apply spectral clustering to topological affinity matrix  $S$ .

```

Algorithm 2 summarizes the complete process of OMELET. The main difference from other ensemble methods lies in applying multiple kernel functions to the CA matrix to expand the diversity of sample similarity relationships, followed by a topological refinement step that eliminates noise and strengthens topological consistency between samples. Additionally, by applying multiple kernels to the CA matrix derived from base clustering results, OMELET achieves its feature-free capability.

3.4. Convergence analysis

In this section, we will theoretically prove the convergence of each subproblem. For convenience, let the objective function Eq. (6) be denoted as $J(\tilde{K}, Z, Y, S, \alpha^{(i)})$. The convergence analysis for each subproblem is as follows. For the $\tilde{K}$ -subproblem, we can compute the first-order and second-order derivatives of J w.r.t. $\tilde{K}$. It is easy to obtain its second-order derivative $\nabla^2 J = 2 > 0$, which indicates that the $\tilde{K}$ -subproblem is a convex problem. Therefore, we have

$$J(\tilde{K}^{t+1}, Z^t, Y^t, S^t, \alpha^{(i)t}) \leq J(\tilde{K}^t, Z^t, Y^t, S^t, \alpha^{(i)t}). \quad (21)$$

For the Z -subproblem, its second-order derivative of J w.r.t. Z is $\nabla^2 J = 2\tilde{K} + 2\lambda I$. According to [37,46], non-negativity constraints do not affect convexity, thus the Z -subproblem is also convex. Therefore, we have

$$J(\tilde{K}^{t+1}, Z^{t+1}, Y^t, S^t, \alpha^{(i)t}) \leq J(\tilde{K}^{t+1}, Z^t, Y^t, S^t, \alpha^{(i)t}). \quad (22)$$

For Eq. (12), we need to compute the Hessian matrix w.r.t. Y to determine its convexity. Therefore, we have

$$\mathbf{H}_Y = \frac{\partial^2 tr(Y^T L_S Y)}{\partial Y \partial Y^T} = 2L_S.$$

It is easy to observe that the Hessian matrix $\mathbf{H}_Y$ is a positive semidefinite matrix, resulting in Eq. (12) being a convex problem. Therefore, we have

$$J(\tilde{K}^{t+1}, Z^{t+1}, Y^{t+1}, S^t, \alpha^{(i)t}) \leq J(\tilde{K}^{t+1}, Z^{t+1}, Y^t, S^t, \alpha^{(i)t}). \quad (23)$$

Algorithm 3 OMELET-SV (The scalable version of OMELET).

Input: CA matrix A , parameter $\lambda, \gamma, p = \lfloor 10\sqrt{n} \rfloor$.

Initialization: $S = Z = I, \alpha^{(i)} = 1/k$, and $\epsilon = 10^{-2}$.

Output: Clustering result C .

- 1: Sampling p anchors from the CA matrix without replacement.
 - 2: Construct a set of anchor kernel matrices $\{K_p^{(1)}, K_p^{(2)}, \dots, K_p^{(k)}\}$ and sample-anchor similarity matrices $\{W_p^{(1)}, W_p^{(2)}, \dots, W_p^{(k)}\}$.
 - 3: Perform Algorithm 2 to obtain the anchor topological affinity matrix S_p and kernel weights α_p .
 - 4: Compute the weighted sum of the sample-anchor similarity matrices using α_p to obtain W .
 - 5: Perform SVD on $S_p, S_p = U_p \Sigma_p V_p^T$. Perform K-means algorithm on R to obtain the clustering result C , where $R = W U_p \Sigma_p^{-\frac{1}{2}}$.
-

The process of updating the variable S is theoretically guaranteed by the ALM method. Moreover, the second derivative of J w.r.t. $\alpha^{(i)}$ can be easily verified to be greater than 0, thus we have

$$J(\tilde{K}^{t+1}, Z^{t+1}, Y^{t+1}, S^{t+1}, \alpha^{(i)t}) \leq J(\tilde{K}^{t+1}, Z^{t+1}, Y^{t+1}, S^t, \alpha^{(i)t}),$$

$$J(\tilde{K}^{t+1}, Z^{t+1}, Y^{t+1}, S^{t+1}, \alpha^{(i)t+1}) \leq J(\tilde{K}^{t+1}, Z^{t+1}, Y^{t+1}, S^{t+1}, \alpha^{(i)t}). \quad (24)$$

Next, we combine Eq. (21)-(24), and obtain

$$J(\tilde{K}^{t+1}, Z^{t+1}, Y^{t+1}, S^{t+1}, \alpha^{(i)t+1}) \leq J(\tilde{K}^t, Z^t, Y^t, S^t, \alpha^{(i)t}).$$

Therefore, OMELET monotonically decreases during each iterative update until convergence. Furthermore, as shown in Fig. 2, the convergence curve obtained from our experiments confirms the convergence of OMELET, which converges within a relatively small number of iterations.

3.5. Complexity analysis

Time complexity: In summary, the proposed OMELET involves updating a total of five variables. For variables $\tilde{K}$, and Z , they either involve matrix multiplication or matrix inverse operations, resulting in a time complexity of $\mathcal{O}(n^3)$. The update for variable S requires updating variable Y first, and their combined complexity is $\mathcal{O}(n^2)$. The overall time complexity of OMELET is $\mathcal{O}(n^3)$. **Space complexity:** For OMELET, the space complexity is mainly determined by storing the CA matrix, k kernel matrices, and the topological affinity graph, all of which are of size $n \times n$. Therefore, the overall space complexity is $\mathcal{O}(kn^2)$.

3.6. Large-scale extension of the OMELET (OMELET-SV)

In this section, we propose a scalable version (OMELET-SV) of the OMELET method to facilitate its application on large-scale datasets. To

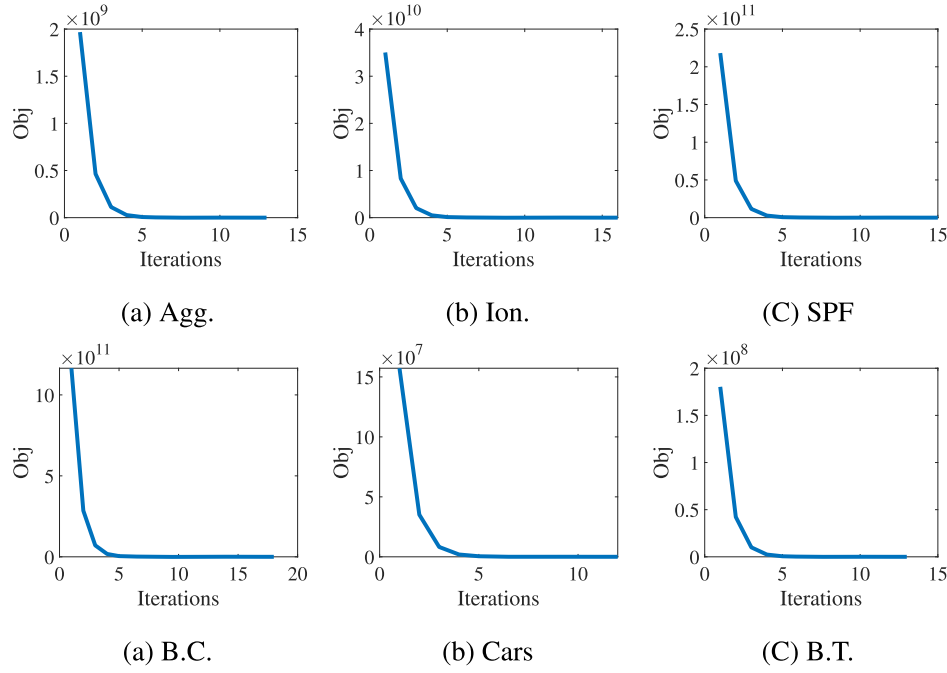


Fig. 2. Convergence curves of the proposed OMELET.

Table 2
Characteristics of datasets.

No.	Dataset	Instance	Feature	Class
D1	ALLAML	72	7129	2
D2	Breast Cancer (B.C.)	683	9	2
D3	Breast Tissue (B.T.)	106	9	6
D4	Aggregation (Agg.)	788	2	7
D5	Cars	392	8	3
D6	Ionosphere (Ion.)	351	34	2
D7	Led7digit	500	7	10
D8	Mammographic (Mam.)	961	5	2
D9	Tr12	313	5804	8
D10	Vertebral3c	310	6	3
D11	FCT	3780	54	7
D12	SPF	1941	27	7
D13	USPS	11,000	256	10

accelerate the efficiency of OMELET, we need to reduce the number of samples in the kernel matrix to overcome computational bottlenecks. Therefore, we select a set of anchors to construct the kernel matrix, which can reduce the time complexity of Algorithm 2 to $\mathcal{O}(p^3)$, where p denotes the number of anchors. In this way, we can obtain a topological affinity matrix of the anchors. However, this only captures the clustering results of the anchor set. To obtain the clustering results for all samples, we need to compute the similarity between the anchors and all samples. To describe the diversity relationship, we still adopt multiple kernel functions to measure the similarity between samples and anchors, and fuse these similarity matrices using the weights of the anchor kernel matrices obtained in Algorithm 2 [34]. Finally, the K-means algorithm is applied to the reconstructed similarity matrix between samples and anchors to obtain the clustering results. The process of OMELET-SV is summarized in Algorithm 3. **Time complexity:** According to the procedure in Algorithm 3, the time complexity of OMELET-SV can be decomposed into several parts. Constructing the anchor kernel matrices and the sample-anchor similarity matrix has time complexity $\mathcal{O}(np^2)$. Running Algorithm 2 to obtain the anchor topological affinity matrix has time complexity $\mathcal{O}(p^3)$. Computing the weighted-sum sample-anchor similarity matrix has time complexity $\mathcal{O}(np)$. Finally, performing the SVD and K-means to obtain the consensus partition has time complexity $\mathcal{O}(p^3 + np)$.

Overall, the total time complexity of OMELET-SV is $\mathcal{O}(np^2 + p^3)$. **Space complexity:** For OMELET-SV, the memory cost comes from k anchor kernel matrices of size $p \times p$ and an anchor-based similarity matrix of size $p \times n$, yielding $\mathcal{O}(kp^2 + pn)$.

4. Experiment

All experiments were conducted on a PC with AMD EPYC 7763 64-Core Processor 2.45 GHz CPU, two NVIDIA GeForce RTX 2080 Ti GPUs and 128 GB memory, running Windows 10 Professional. The algorithms were implemented in MATLAB (R2023b). The code is available at <https://github.com/middle258/OMELET>.

4.1. Experiment setup

We conduct extensive experiments on 13 real datasets,¹ with detailed information provided in Table 2. To set up fair comparisons, we collect publicly available base clustering results released by prior works [10,12,47]. To obtain the base clustering result labels of other datasets, we randomly run the K-means algorithm 100 times. The cluster parameter for the K-means algorithm is set to $[c, \min(\sqrt{n}, 100)]$ [48], where c and n denote the number of the classes and samples in the dataset, respectively. Then, we set the ensemble size $m = 20$ for all algorithms, repeating the experiments 10 times and evaluating the performance using the mean and standard deviation. To ensure a fair comparison, the base clustering sets used by different methods are kept identical. To demonstrate the effectiveness of the ensemble strategy, we report the average (KM-avg) and best results (KM-best) over the base clustering results. We compare against 17 representative ensemble clustering algorithms and 1 multiple kernel clustering algorithm, which are briefly introduced as follows: •PTA-AL, PTA-CL, PTGP (TKDE, 2016) [49]. They are three ensemble methods that introduce the microcluster theory. Their differences lie in using hierarchical agglomerative clustering (PTA-AL), complete-link hierarchical clustering (PTA-CL), and Tcut (PTGP) to obtain the final results. •RSEC-Z, RSEC-H (TKDD, 2019)

¹ <http://archive.ics.uci.edu/datasets>

Table 3Clustering performance measured by ARI on D1–D10 (mean $\pm$ std %).

Method	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
KM-avg	10.0 $\pm$ 6.6	43.5 $\pm$ 21.6	15.8 $\pm$ 3.1	43.8 $\pm$ 15.1	3.8 $\pm$ 2.2	18.5 $\pm$ 4.4	40.6 $\pm$ 4.0	4.9 $\pm$ 2.6	21.8 $\pm$ 8.4	21.6 $\pm$ 6.4
KM-best	23.8	85.2	20.5	76.4	9.2	33.6	49.3	13.7	44.1	32.5
PTA-AL	12.2 $\pm$ 5.6	82.5 $\pm$ 21.0	12.9 $\pm$ 3.9	75.5 $\pm$ 11.5	16.4 $\pm$ 4.1	7.0 $\pm$ 13.7	38.4 $\pm$ 3.4	12.8 $\pm$ 4.9	30.2 $\pm$ 7.0	29.1 $\pm$ 19.2
PTA-CL	10.7 $\pm$ 2.7	88.2 $\pm$ 2.3	10.1 $\pm$ 5.9	71.7 $\pm$ 11.4	12.0 $\pm$ 5.1	12.2 $\pm$ 6.9	38.8 $\pm$ 3.0	7.6 $\pm$ 6.6	30.1 $\pm$ 6.2	19.2 $\pm$ 14.8
PTGP	13.2 $\pm$ 3.8	88.3 $\pm$ 0.9	14.8 $\pm$ 4.1	75.7 $\pm$ 7.5	14.3 $\pm$ 3.7	21.4 $\pm$ 11.2	43.8 $\pm$ 3.1	14.5 $\pm$ 1.4	29.9 $\pm$ 6.4	34.5 $\pm$ 4.1
RSEC-Z	8.4 $\pm$ 4.5	51.8 $\pm$ 46.1	12.7 $\pm$ 6.1	90.4 $\pm$ 5.2	11.4 $\pm$ 6.7	10.8 $\pm$ 9.1	36.5 $\pm$ 20.9	7.4 $\pm$ 7.4	19.2 $\pm$ 10.7	23.7 $\pm$ 18.6
RSEC-H	8.4 $\pm$ 11.1	61.4 $\pm$ 43.2	9.0 $\pm$ 9.6	97.9 $\pm$ 2.9	12.9 $\pm$ 6.9	12.5 $\pm$ 9.8	35.6 $\pm$ 23.4	7.0 $\pm$ 6.8	19.1 $\pm$ 13.3	27.1 $\pm$ 22.9
LWEA	12.0 $\pm$ 4.6	82.2 $\pm$ 2.7	11.0 $\pm$ 3.3	94.4 $\pm$ 6.4	12.1 $\pm$ 5.2	21.1 $\pm$ 10.7	43.7 $\pm$ 2.5	13.1 $\pm$ 1.0	35.8 $\pm$ 15.1	23.0 $\pm$ 18.0
LGWP	10.1 $\pm$ 2.4	87.3 $\pm$ 0.9	17.3 $\pm$ 4.0	96.9 $\pm$ 1.0	13.5 $\pm$ 1.0	17.8 $\pm$ 0.2	49.8 $\pm$ 2.6	13.9 $\pm$ 1.6	31.3 $\pm$ 12.5	21.0 $\pm$ 22.8
ECPCS-HC	11.9 $\pm$ 5.2	86.4 $\pm$ 3.0	12.4 $\pm$ 3.8	98.0 $\pm$ 1.6	9.0 $\pm$ 7.6	16.1 $\pm$ 1.6	40.0 $\pm$ 4.4	11.5 $\pm$ 4.3	14.6 $\pm$ 1.8	1.8 $\pm$ 0.4
ECPCS-MC	10.4 $\pm$ 2.3	86.7 $\pm$ 0.8	15.8 $\pm$ 2.3	92.4 $\pm$ 4.6	12.5 $\pm$ 2.4	17.7 $\pm$ 1.3	43.8 $\pm$ 2.6	1.4 $\pm$ 4.3	18.9 $\pm$ 3.0	2.9 $\pm$ 0.4
DREC	12.8 $\pm$ 4.3	82.8 $\pm$ 0.9	10.1 $\pm$ 1.7	71.5 $\pm$ 16.9	6.6 $\pm$ 7.4	17.4 $\pm$ 1.1	43.5 $\pm$ 3.8	12.8 $\pm$ 1.6	26.1 $\pm$ 4.5	33.8 $\pm$ 15.9
SCCBG	15.1 $\pm$ 5.9	89.0 $\pm$ 0.4	14.9 $\pm$ 3.8	98.1 $\pm$ 2.8	12.2 $\pm$ 4.9	17.6 $\pm$ 0.5	40.2 $\pm$ 6.8	12.6 $\pm$ 1.8	16.8 $\pm$ 1.8	2.3 $\pm$ 0.3
SPCE	14.8 $\pm$ 4.5	84.6 $\pm$ 1.7	18.7 $\pm$ 1.1	99.0 $\pm$ 0.1	9.6 $\pm$ 7.6	24.8 $\pm$ 3.4	38.8 $\pm$ 1.7	9.9 $\pm$ 3.5	27.2 $\pm$ 5.3	28.5 $\pm$ 4.1
CESHL	10.2 $\pm$ 2.1	87.0 $\pm$ 1.8	11.7 $\pm$ 3.6	96.9 $\pm$ 3.9	13.0 $\pm$ 2.7	17.9 $\pm$ 0.4	45.4 $\pm$ 5.3	13.7 $\pm$ 1.2	19.2 $\pm$ 3.5	8.3 $\pm$ 17.7
TRCE	18.0 $\pm$ 7.0	88.9 $\pm$ 0.5	17.3 $\pm$ 3.3	98.1 $\pm$ 2.6	18.6 $\pm$ 6.9	18.3 $\pm$ 2.5	47.8 $\pm$ 2.2	13.3 $\pm$ 1.4	22.8 $\pm$ 4.7	9.3 $\pm$ 16.0
TRCE-K	18.8 $\pm$ 6.4	88.4 $\pm$ 1.4	11.2 $\pm$ 2.8	97.2 $\pm$ 3.7	17.4 $\pm$ 4.9	17.4 $\pm$ 1.0	44.7 $\pm$ 3.4	14.0 $\pm$ 1.1	15.0 $\pm$ 2.9	30.9 $\pm$ 12.0
CEAM	7.5 $\pm$ 6.3	82.1 $\pm$ 9.8	18.5 $\pm$ 0.8	64.0 $\pm$ 5.0	6.2 $\pm$ 7.6	11.4 $\pm$ 6.8	42.2 $\pm$ 4.5	10.1 $\pm$ 6.7	32.5 $\pm$ 4.5	23.7 $\pm$ 8.9
FSEC	10.1 $\pm$ 2.4	87.6 $\pm$ 0.7	16.5 $\pm$ 0.9	85.5 $\pm$ 9.1	12.0 $\pm$ 2.1	17.3 $\pm$ 0.7	46.8 $\pm$ 3.1	13.9 $\pm$ 0.9	30.2 $\pm$ 4.0	33.0 $\pm$ 1.8
SMKKM	17.6 $\pm$ 5.7	66.9 $\pm$ 24.3	19.2 $\pm$ 1.1	73.3 $\pm$ 4.4	26.1 $\pm$ 3.1	12.3 $\pm$ 2.3	36.5 $\pm$ 1.7	5.2 $\pm$ 5.6	33.0 $\pm$ 2.3	35.4 $\pm$ 13.1
OMELET	19.6 $\pm$ 6.1	89.2 $\pm$ 0.3	24.7 $\pm$ 1.3	99.0 $\pm$ 0.0	27.8 $\pm$ 9.0	45.7 $\pm$ 8.0	50.5 $\pm$ 0.7	14.5 $\pm$ 10.2	37.7 $\pm$ 4.7	52.9 $\pm$ 11.4

[50]. They are both robust spectral ensemble clustering method, capable of simultaneously learning the low-rank structure Z and the consensus partition H . •**LWEA**, **LGWP** (TCYB, 2018) [10]. They use the same entropy-weighted approach with different partitioning methods. •**ECPCS-HC**, **ECPCS-MC** (TSMC, 2021) [48]. They both enhance the CA matrix by propagating cluster-wise similarities, but with different partitioning methods. •**DREC** (Neurocomputing, 2019) [47]. It utilizes dense representation and microclusters to obtain consensus results. •**SCCBG** (IJCAI, 2021) [12]. It applies a self-paced learning framework to a bipartite graph. •**SPCE** (TNNLS, 2021) [11]. It retains the most reliable results based on the self-paced learning framework. •**CESHL** (Information Fusion, 2022) [20]. It applies hypergraph learning to obtain the consensus result. •**TRCE**, **TRCE-K** (AAAI, 2021) [51]. It considers the robustness of ensemble clustering from three levels. TRCE-K is a variant of TRCE that takes the same CA-derived kernel matrices as OMELET as input. •**CEAM** (TKDE, 2024) [52]. It is an ensemble clustering method based on adaptive multiplexing. •**FSEC** (Information Fusion, 2025) [53]. It achieves ensemble clustering by selecting representative samples (anchors). •**SMKKM** (TPAMI, 2023) [33]. It extends the widely used supervised kernel alignment criterion to multiple kernel clustering. It uses the same kernel matrices as OMELET as input.

For all baselines, we use implementations released by the original authors and tune their hyper-parameters via grid search within the parameter ranges reported in the corresponding papers. We report the best results obtained from these searches, ensuring that each baseline is evaluated under its strongest settings. To generate the base kernels, we employ three types of kernel functions:

$$\text{Gaussian kernel : } \mathbb{K}(x_i, x_j) = \exp\left(\frac{-\|x_i - x_j\|^2}{r^2}\right)$$

$$\text{Linear kernel : } \mathbb{K}(x_i, x_j) = x_i^\top x_j$$

$$\text{Polynomial kernel : } \mathbb{K}(x_i, x_j) = (x_i^\top x_j + 1)^b. \quad (25)$$

In our experiments, we use two Gaussian kernels, one polynomial kernel, and one linear kernel. To enhance the diversity of the Gaussian kernels, we set their parameters as $r^2 = \sigma$ and $r^2 = \sigma^2$, where σ ranges from $[2^1, 2^2, \dots, 2^5, 2^6]$. We set the polynomial kernel parameter $b = 2$. We evaluate the experimental results from different perspectives using three popular metrics, including Adjusted Rand Index (ARI), F1-score, and Normalized Mutual Information (NMI). The larger their values, the better the clustering performance.

4.2. Experiment results

Tables 3, 4, 5, and 6 report the results of all methods on the three metrics. We observe that OMELET exhibits superior performance across three metrics. OMELET surpasses the average results of the base clustering result set across all datasets and metrics, thereby demonstrating the effectiveness of this ensemble strategy. However, in a very small number of datasets, OMELET does not surpass KM-best in all metrics. Possible reasons include: (i) The optimal base clustering results are not included in the ensemble members; (ii) The optimal base clustering results are included in the ensemble members, but reduce the diversity of the base clustering result set, leading to decreased performance. Note that the OMELET method still performs well on the larger-scale dataset USPS, indicating that OMELET has good scalability. For the ARI metric, OMELET achieves optimal results across all datasets. However, in the NMI and F1-score metrics, OMELET fails to achieve optimal results 3 - 4 times, respectively. OMELET also outperforms TRCE-K and SMKKM, both of which take the same CA-derived base kernels as input. Compared with TRCE, TRCE-K achieves higher performance on some datasets, especially on D10, supporting the effectiveness of multiple kernel learning. However, SMKKM and TRCE-K do not mitigate the potential noise introduced by multiple kernel learning, which results in degraded performance. In contrast, OMELET incorporates topological-aware similarity learning to alleviate this issue.

Competitive methods include LWEA, TRCE, PTGP, and SPCE. Among them, the first three achieve optimal results twice across the three evaluation metrics, whereas SPCE achieves optimal results 4 times. We provide the following possible explanations: (i) The self-paced learning framework excels at evaluating the reliability of similarity relationships among samples [11]. SPCE further expands the scope of reliable information, preserving sample information from both same-class and different-class samples as recognized by all base clustering results. (ii) OMELET mitigates noise through topological similarity learning, while TRCE addresses noise from three different perspectives, making it potentially more suitable for datasets heavily affected by noise. (iii) In certain datasets, the CA matrix may contain many low-quality similarity relationships. The diverse similarities learned via consensus kernel learning can exacerbate this issue, thereby degrading the quality of the topological affinity matrix.

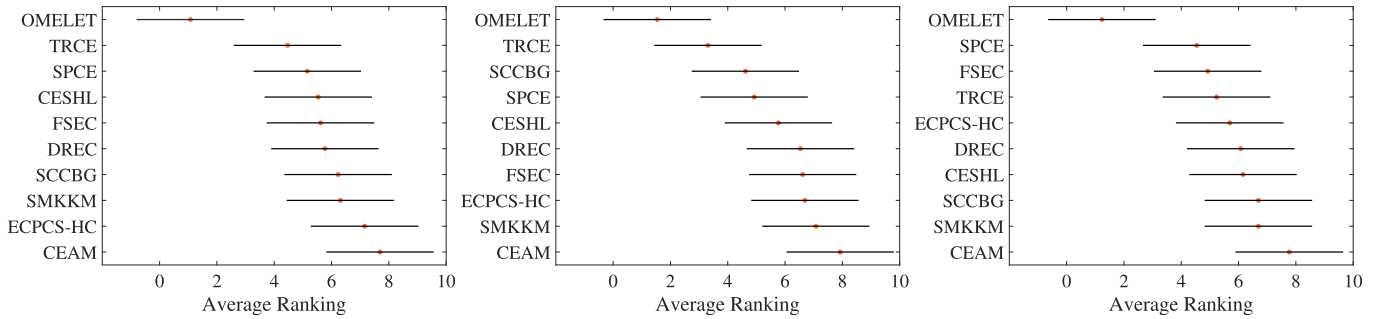
We conduct a Friedman test [54] on three evaluation metrics to assess the significance of the performance differences. As shown in Fig. 3, the orange dots denote the average ranks. If the confidence intervals

Table 4Clustering performance measured by F1-score on D1–D10 (mean $\pm$ std %).

Method	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
KM-avg	56.6 $\pm$ 3.2	60.2 $\pm$ 19.1	33.9 $\pm$ 1.7	49.5 $\pm$ 15.2	21.6 $\pm$ 9.0	48.2 $\pm$ 9.2	46.5 $\pm$ 3.3	23.5 $\pm$ 12.4	36.7 $\pm$ 5.5	40.1 $\pm$ 11.1
KM-best	63.9	93.3	36.4	81.0	44.8	63.3	54.5	57.5	52.5	61.5
PTA-AL	60.9 $\pm$ 4.7	91.9 $\pm$ 9.8	32.6 $\pm$ 2.1	81.3 $\pm$ 8.9	55.8 $\pm$ 4.1	61.4 $\pm$ 5.9	44.4 $\pm$ 3.7	58.8 $\pm$ 1.6	42.7 $\pm$ 4.9	59.1 $\pm$ 7.7
PTA-CL	60.5 $\pm$ 4.8	94.6 $\pm$ 1.0	31.0 $\pm$ 3.5	78.4 $\pm$ 9.2	53.6 $\pm$ 2.3	59.1 $\pm$ 2.3	44.9 $\pm$ 3.2	56.8 $\pm$ 2.5	42.4 $\pm$ 4.4	53.1 $\pm$ 4.3
PTGP	70.2 $\pm$ 0.0	94.6 $\pm$ 0.4	33.4 $\pm$ 2.2	81.8 $\pm$ 5.3	53.6 $\pm$ 2.0	62.6 $\pm$ 6.2	49.7 $\pm$ 2.6	58.0 $\pm$ 0.3	42.1 $\pm$ 4.4	59.2 $\pm$ 2.7
RSEC-Z	61.1 $\pm$ 5.1	80.4 $\pm$ 17.9	32.4 $\pm$ 6.5	92.4 $\pm$ 4.3	62.3 $\pm$ 1.5	68.1 $\pm$ 1.2	44.4 $\pm$ 16.2	66.0 $\pm$ 0.4	36.9 $\pm$ 7.1	60.0 $\pm$ 9.7
RSEC-H	60.6 $\pm$ 4.5	84.1 $\pm$ 16.9	31.2 $\pm$ 5.6	98.4 $\pm$ 2.3	61.0 $\pm$ 1.9	67.7 $\pm$ 1.5	44.0 $\pm$ 18.0	65.3 $\pm$ 0.8	38.9 $\pm$ 7.6	61.6 $\pm$ 7.1
LWEA	57.5 $\pm$ 2.4	92.0 $\pm$ 1.2	31.8 $\pm$ 2.1	95.5 $\pm$ 5.1	51.2 $\pm$ 4.9	62.2 $\pm$ 6.0	50.2 $\pm$ 2.1	62.9 $\pm$ 4.1	48.9 $\pm$ 10.6	57.8 $\pm$ 13.0
LWGP	56.5 $\pm$ 1.6	94.2 $\pm$ 0.4	33.8 $\pm$ 2.6	97.5 $\pm$ 0.8	51.1 $\pm$ 3.9	68.1 $\pm$ 4.8	54.9 $\pm$ 2.3	65.7 $\pm$ 2.7	46.5 $\pm$ 7.1	57.7 $\pm$ 10.8
ECPCS-HC	57.5 $\pm$ 2.7	93.8 $\pm$ 1.3	32.6 $\pm$ 2.4	98.5 $\pm$ 1.2	48.3 $\pm$ 2.6	59.6 $\pm$ 0.8	47.3 $\pm$ 3.5	58.7 $\pm$ 2.8	35.3 $\pm$ 1.7	49.1 $\pm$ 0.1
ECPCS-MC	56.7 $\pm$ 1.2	93.9 $\pm$ 0.3	34.7 $\pm$ 1.4	93.9 $\pm$ 3.7	48.5 $\pm$ 1.4	60.5 $\pm$ 0.6	49.8 $\pm$ 2.2	65.7 $\pm$ 3.0	36.8 $\pm$ 1.6	49.4 $\pm$ 0.1
DREC	57.9 $\pm$ 2.3	92.3 $\pm$ 0.4	31.1 $\pm$ 0.7	77.1 $\pm$ 13.7	45.6 $\pm$ 2.6	60.3 $\pm$ 0.6	49.8 $\pm$ 3.1	57.0 $\pm$ 0.7	40.1 $\pm$ 2.9	59.4 $\pm$ 6.9
SCCBG	60.0 $\pm$ 4.3	94.9 $\pm$ 0.2	33.8 $\pm$ 2.6	98.5 $\pm$ 2.2	56.6 $\pm$ 4.9	70.0 $\pm$ 0.0	47.3 $\pm$ 5.4	66.6 $\pm$ 0.0	36.6 $\pm$ 1.4	52.6 $\pm$ 1.9
SPCE	60.1 $\pm$ 5.1	93.0 $\pm$ 0.7	36.2 $\pm$ 1.3	99.2 $\pm$ 0.1	55.2 $\pm$ 6.0	61.7 $\pm$ 1.4	42.8 $\pm$ 1.4	66.4 $\pm$ 0.5	36.6 $\pm$ 4.0	53.8 $\pm$ 3.1
CESHL	56.5 $\pm$ 1.1	94.1 $\pm$ 0.8	32.2 $\pm$ 2.3	97.5 $\pm$ 3.1	49.5 $\pm$ 0.5	60.5 $\pm$ 0.2	51.6 $\pm$ 4.1	58.7 $\pm$ 1.6	36.9 $\pm$ 2.0	52.1 $\pm$ 8.5
TRCE	70.2 $\pm$ 0.0	94.9 $\pm$ 0.2	35.8 $\pm$ 2.6	98.5 $\pm$ 2.1	63.3 $\pm$ 1.0	70.0 $\pm$ 0.0	53.2 $\pm$ 1.9	66.7 $\pm$ 0.0	38.8 $\pm$ 3.5	56.1 $\pm$ 4.5
TRCE-K	70.2 $\pm$ 0.0	94.7 $\pm$ 0.6	31.8 $\pm$ 1.6	97.8 $\pm$ 3.0	62.8 $\pm$ 0.2	70.0 $\pm$ 0.0	50.5 $\pm$ 2.9	66.7 $\pm$ 0.0	34.2 $\pm$ 1.9	59.0 $\pm$ 4.8
CEAM	56.2 $\pm$ 0.0	91.8 $\pm$ 5.5	35.5 $\pm$ 14.3	70.9 $\pm$ 26.2	44.4 $\pm$ 4.6	57.6 $\pm$ 0.6	48.2 $\pm$ 8.4	56.3 $\pm$ 0.8	42.7 $\pm$ 2.4	51.7 $\pm$ 1.2
FSEC	56.5 $\pm$ 1.2	94.3 $\pm$ 0.3	34.5 $\pm$ 0.9	88.4 $\pm$ 7.4	47.8 $\pm$ 1.3	60.2 $\pm$ 0.4	52.2 $\pm$ 2.7	57.3 $\pm$ 0.2	42.6 $\pm$ 3.0	58.6 $\pm$ 1.1
SMKKM	60.7 $\pm$ 3.1	84.5 $\pm$ 11.3	35.4 $\pm$ 1.6	78.4 $\pm$ 3.6	56.2 $\pm$ 2.8	57.9 $\pm$ 0.8	43.3 $\pm$ 1.5	52.9 $\pm$ 2.9	43.2 $\pm$ 2.0	59.1 $\pm$ 8.7
OMELET	67.8 $\pm$ 1.7	95.0 $\pm$ 0.2	41.6 $\pm$ 0.7	99.2 $\pm$ 0.0	64.2 $\pm$ 2.6	76.6 $\pm$ 4.4	55.5 $\pm$ 0.6	65.4 $\pm$ 0.4	47.6 $\pm$ 3.6	70.6 $\pm$ 7.3

Table 5Clustering performance measured by NMI on D1–D10 (mean $\pm$ std %).

Method	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
KM-avg	9.1 $\pm$ 5.0	47.1 $\pm$ 11.3	35.5 $\pm$ 3.4	75.6 $\pm$ 5.5	16.9 $\pm$ 1.9	21.2 $\pm$ 4.3	54.7 $\pm$ 2.7	10.4 $\pm$ 1.2	47.5 $\pm$ 6.6	38.6 $\pm$ 2.4
KM-best	28.7	75.5	43.6	87.3	21.1	32.5	61.5	13.0	64.9	43.5
PTA-AL	12.6 $\pm$ 1.6	77.0 $\pm$ 14.9	27.0 $\pm$ 3.7	82.4 $\pm$ 4.4	20.3 $\pm$ 0.9	8.6 $\pm$ 7.8	51.8 $\pm$ 1.6	10.7 $\pm$ 2.6	51.9 $\pm$ 5.7	37.6 $\pm$ 19.3
PTA-CL	12.2 $\pm$ 1.0	80.2 $\pm$ 3.5	23.6 $\pm$ 5.7	78.8 $\pm$ 6.0	16.4 $\pm$ 2.1	10.6 $\pm$ 2.3	51.8 $\pm$ 1.5	6.5 $\pm$ 4.9	51.5 $\pm$ 4.4	30.2 $\pm$ 15.1
PTGP	12.5 $\pm$ 1.4	80.0 $\pm$ 1.7	29.9 $\pm$ 4.3	80.3 $\pm$ 4.2	20.4 $\pm$ 0.7	15.9 $\pm$ 8.6	55.4 $\pm$ 1.7	11.1 $\pm$ 0.8	51.7 $\pm$ 4.5	46.3 $\pm$ 5.5
RSEC-Z	7.8 $\pm$ 3.0	55.0 $\pm$ 32.3	27.4 $\pm$ 6.4	92.5 $\pm$ 2.4	22.0 $\pm$ 1.6	10.6 $\pm$ 3.3	50.9 $\pm$ 13.6	7.3 $\pm$ 4.3	46.2 $\pm$ 10.2	30.5 $\pm$ 24.3
RSEC-H	7.4 $\pm$ 3.1	62.3 $\pm$ 30.6	23.2 $\pm$ 11.6	97.7 $\pm$ 2.2	22.0 $\pm$ 1.0	11.5 $\pm$ 3.6	49.8 $\pm$ 18.2	7.0 $\pm$ 3.9	47.1 $\pm$ 6.9	32.1 $\pm$ 25.1
LWEA	9.8 $\pm$ 2.6	71.4 $\pm$ 3.7	24.9 $\pm$ 2.3	94.9 $\pm$ 4.7	20.6 $\pm$ 0.9	15.6 $\pm$ 8.4	53.2 $\pm$ 1.9	10.6 $\pm$ 0.4	48.6 $\pm$ 5.5	37.1 $\pm$ 17.4
LWGP	8.7 $\pm$ 4.6	78.1 $\pm$ 1.3	29.2 $\pm$ 3.1	96.0 $\pm$ 1.1	19.8 $\pm$ 0.8	13.2 $\pm$ 0.3	58.5 $\pm$ 1.7	10.7 $\pm$ 0.9	44.8 $\pm$ 7.3	28.1 $\pm$ 12.9
ECPCS-HC	10.1 $\pm$ 3.0	78.2 $\pm$ 4.9	31.0 $\pm$ 2.6	97.7 $\pm$ 1.4	20.7 $\pm$ 0.4	12.5 $\pm$ 1.0	54.1 $\pm$ 1.8	9.6 $\pm$ 3.3	41.3 $\pm$ 2.0	22.1 $\pm$ 0.5
ECPCS-MC	9.2 $\pm$ 1.3	77.4 $\pm$ 1.2	33.6 $\pm$ 1.5	93.7 $\pm$ 3.0	20.7 $\pm$ 0.3	13.2 $\pm$ 1.0	55.1 $\pm$ 1.8	1.3 $\pm$ 3.3	43.1 $\pm$ 2.3	23.2 $\pm$ 0.4
DREC	10.3 $\pm$ 2.6	72.0 $\pm$ 1.2	24.3 $\pm$ 1.6	82.1 $\pm$ 9.6	17.8 $\pm$ 3.8	12.8 $\pm$ 0.7	54.0 $\pm$ 2.9	10.0 $\pm$ 1.0	46.9 $\pm$ 4.0	47.8 $\pm$ 13.2
SCCBG	12.6 $\pm$ 3.3	81.0 $\pm$ 1.0	31.7 $\pm$ 6.5	97.4 $\pm$ 2.8	20.4 $\pm$ 0.9	13.3 $\pm$ 0.3	51.5 $\pm$ 5.3	10.6 $\pm$ 0.6	37.3 $\pm$ 4.1	17.1 $\pm$ 0.3
SPCE	13.5 $\pm$ 1.4	74.4 $\pm$ 2.5	38.0 $\pm$ 1.5	98.3 $\pm$ 0.2	20.2 $\pm$ 0.8	20.0 $\pm$ 2.2	50.4 $\pm$ 0.8	9.9 $\pm$ 0.9	46.5 $\pm$ 2.1	38.3 $\pm$ 4.8
CESHL	8.8 $\pm$ 1.2	77.8 $\pm$ 2.8	25.3 $\pm$ 2.5	96.2 $\pm$ 3.8	19.9 $\pm$ 0.6	13.4 $\pm$ 0.3	55.0 $\pm$ 4.1	10.8 $\pm$ 0.4	40.7 $\pm$ 3.9	20.6 $\pm$ 10.1
TRCE	13.3 $\pm$ 4.1	81.2 $\pm$ 0.8	30.1 $\pm$ 2.7	97.5 $\pm$ 2.5	19.8 $\pm$ 0.7	13.9 $\pm$ 2.5	57.5 $\pm$ 1.4	10.6 $\pm$ 0.4	45.4 $\pm$ 3.1	22.6 $\pm$ 12.2
TRCE-K	13.6 $\pm$ 3.9	80.4 $\pm$ 2.4	25.9 $\pm$ 2.6	96.8 $\pm$ 3.3	20.0 $\pm$ 0.5	13.0 $\pm$ 0.5	57.7 $\pm$ 1.3	10.7 $\pm$ 0.4	39.8 $\pm$ 3.5	41.0 $\pm$ 9.1
CEAM	8.3 $\pm$ 5.8	72.7 $\pm$ 12.3	33.5 $\pm$ 1.4	78.6 $\pm$ 1.6	13.5 $\pm$ 5.1	10.1 $\pm$ 4.9	55.4 $\pm$ 2.4	7.8 $\pm$ 5.1	48.9 $\pm$ 4.0	32.5 $\pm$ 9.6
FSEC	9.0 $\pm$ 1.4	78.8 $\pm$ 1.3	33.8 $\pm$ 1.4	90.8 $\pm$ 4.5	20.7 $\pm$ 0.4	13.4 $\pm$ 0.4	56.9 $\pm$ 2.1	10.8 $\pm$ 0.3	52.2 $\pm$ 2.9	46.8 $\pm$ 1.8
SMKKM	13.4 $\pm$ 4.1	65.0 $\pm$ 17.9	36.1 $\pm$ 1.6	85.5 $\pm$ 1.9	20.8 $\pm$ 2.2	10.7 $\pm$ 0.9	54.1 $\pm$ 0.9	4.0 $\pm$ 4.3	52.6 $\pm$ 3.7	38.9 $\pm$ 10.3
OMELET	14.8 $\pm$ 2.9	82.7 $\pm$ 0.3	40.5 $\pm$ 1.4	98.5 $\pm$ 0.0	22.8 $\pm$ 0.4	36.4 $\pm$ 7.0	59.3 $\pm$ 1.1	11.2 $\pm$ 0.6	56.3 $\pm$ 3.7	47.1 $\pm$ 4.3



(a) ARI

(b) F1-score

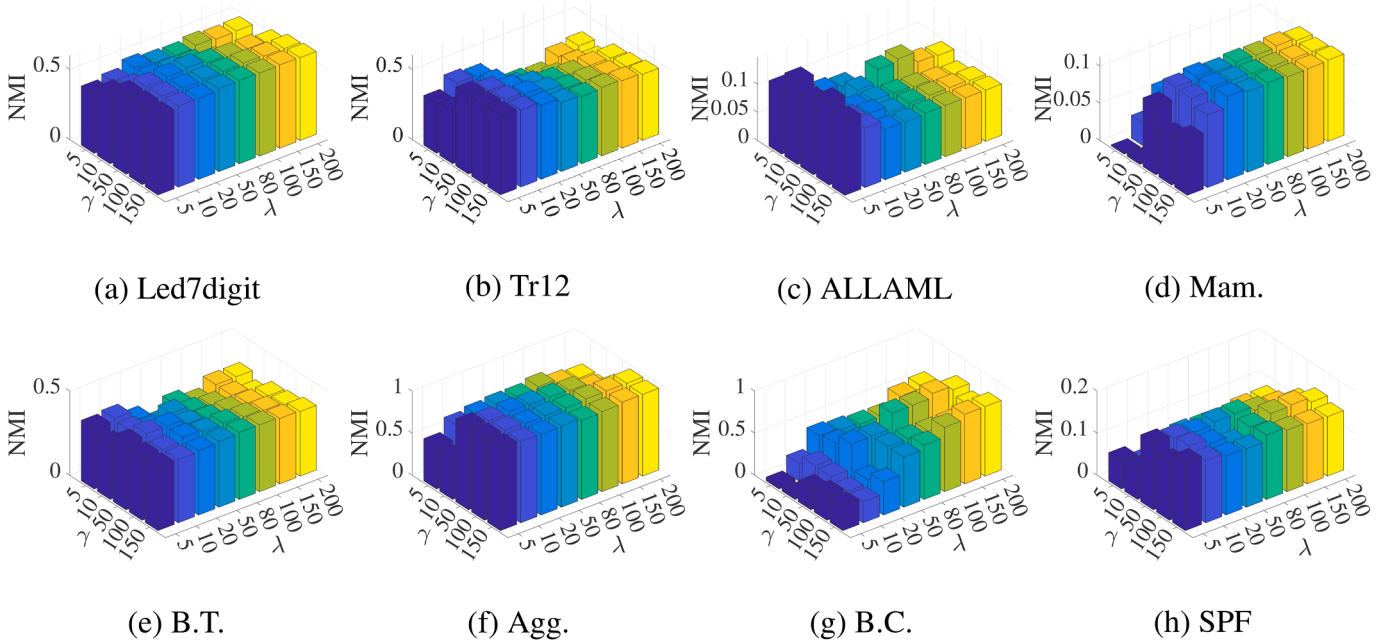
(c) NMI

Fig. 3. Friedman test in terms of ARI, F1-score and NMI.

Table 6

Clustering performance on D11–D13 measured by ARI, F1-score, and NMI (mean±std %).

Method	D11			D12			D13		
	ARI	F1	NMI	ARI	F1	NMI	ARI	F1	NMI
KM-avg	7.5±1.7	14.0±5.7	25.5±3.3	4.4±1.3	14.6±6.4	15.7±2.9	21.5±6.6	24.9±7.9	53.5±5.5
KM-best	12.0	29.9	29.0	7.3	34.8	18.0	33.9	39.9	57.8
PTA-AL	13.0±2.7	26.7±2.5	22.4±2.8	8.9±1.3	26.4±1.8	15.5±1.5	52.1±6.5	57.3±5.7	63.5±4.3
PTA-CL	12.1±1.8	25.4±1.4	21.1±2.5	8.1±1.9	25.4±1.7	13.3±2.4	37.0±3.3	43.9±2.9	53.6±2.1
PTGP	13.0±1.6	25.7±1.5	24.0±2.4	9.5±1.5	26.1±1.4	15.8±1.3	49.4±2.8	54.8±2.4	62.2±1.6
RSEC-Z	10.0±2.5	25.2±2.1	19.7±3.0	5.3±2.3	27.0±1.0	9.9±3.4	42.6±2.3	49.5±2.0	61.6±1.4
RSEC-H	9.4±3.0	25.1±2.5	19.4±3.8	6.0±2.5	28.1±1.4	10.9±4.1	43.4±2.4	50.1±2.0	61.5±1.2
LWEA	12.3±1.4	25.8±1.4	21.6±1.8	9.2±1.8	29.5±1.7	17.8±2.2	53.7±5.0	58.9±4.2	65.1±3.7
LWGP	10.0±1.1	23.9±1.7	19.1±1.9	9.6±1.5	29.3±1.4	17.5±1.3	47.8±1.3	53.8±1.1	63.3±1.2
ECPCS-HC	14.0±2.5	28.0±3.0	23.3±2.7	6.7±1.5	27.9±0.9	14.3±1.8	44.4±2.4	51.2±2.1	61.7±1.6
ECPCS-MC	11.2±1.3	24.9±0.9	20.9±1.6	6.7±0.5	27.7±0.6	14.3±1.1	47.2±1.8	53.0±1.6	62.8±1.8
DREC	12.7±2.5	25.5±2.0	23.0±3.5	8.9±1.4	27.0±1.3	15.2±1.0	52.8±6.6	57.7±5.7	63.9±3.2
SCCBG	10.0±3.4	28.4±1.6	16.4±3.2	6.2±0.4	35.5±0.6	13.1±1.0	45.1±2.6	51.9±2.6	59.0±2.2
SPCE	12.1±3.4	27.7±2.8	27.2±1.5	10.1±2.1	38.0±5.2	20.2±1.5	42.5±3.8	48.9±2.8	56.8±2.6
CESHL	12.1±2.7	28.7±1.5	20.8±2.9	7.6±1.7	33.4±2.6	15.1±2.4	45.7±2.2	52.0±2.1	60.9±1.6
TRCE	10.0±3.5	26.6±1.6	18.9±3.2	7.6±1.7	36.4±2.1	13.8±2.1	47.0±2.3	53.3±2.1	61.6±2.0
TRCE-K	8.94±3.4	27.2±1.5	16.5±3.7	5.0±2.1	36.3±0.0	8.8±3.0	30.2±4.7	38.2±4.0	48.3±3.4
CEAM	12.0±2.6	27.0±0.0	19.3±2.6	6.7±1.8	26.2±0.0	11.3±3.2	35.5±4.1	42.2±2.2	51.9±3.9
FSEC	10.1±0.7	23.7±0.7	20.0±0.9	7.1±0.5	26.2±0.9	13.6±0.6	42.8±3.8	48.8±3.5	58.8±2.8
SMKKM	10.5±1.0	25.0±1.0	18.3±1.6	7.1±1.7	25.5±1.9	9.9±1.4	26.3±4.5	34.6±3.8	43.4±4.2
OMELET	14.1±2.2	29.2±1.7	23.6±2.6	11.3±2.7	35.8±0.3	16.3±1.7	54.1±5.1	58.9±4.5	66.3±3.9

**Fig. 4.** NMI w.r.t. γ and λ .

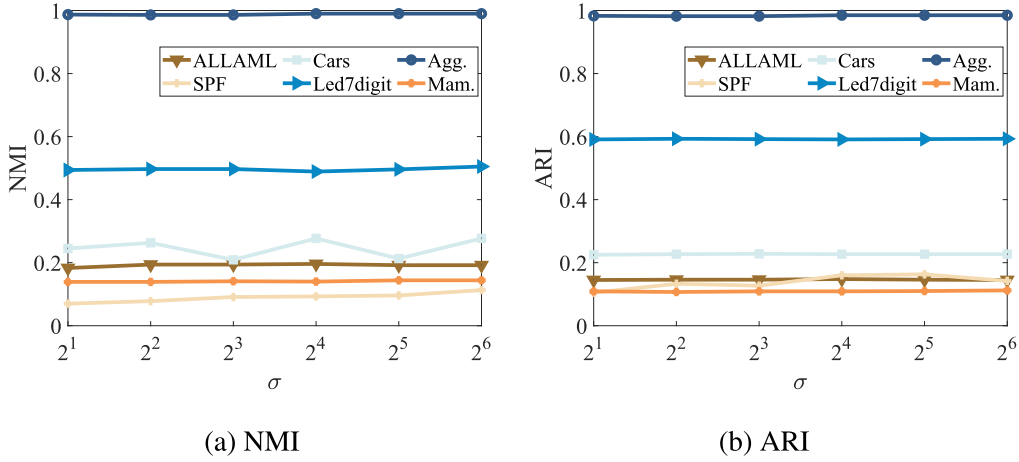
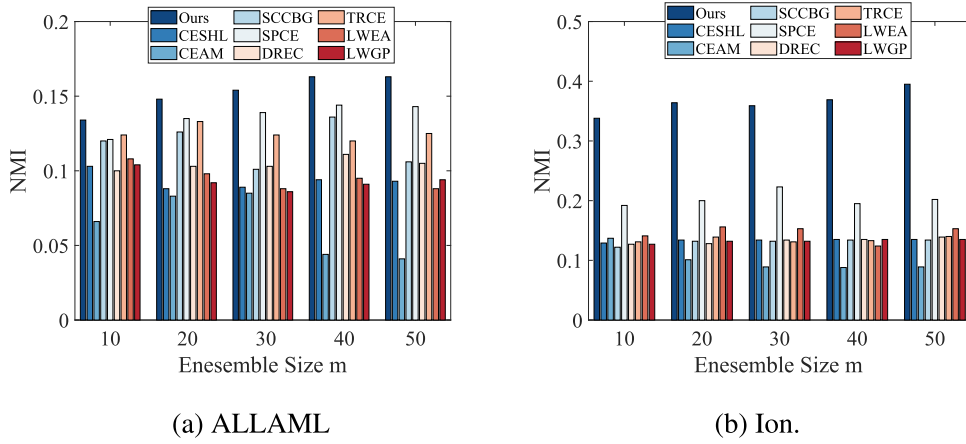
of two methods do not overlap, it indicates a statistically significant difference between them. We observe that OMELET achieves the best (i.e., lowest) average rank on all three metrics and shows statistically significant improvements over most competing methods. In particular, on ARI, OMELET outperforms all other methods except TRCE. Overall, these statistical results provide additional evidence for the effectiveness of OMELET.

4.3. Parameter sensitivity analysis

The sensitivity results of the proposed OMELET are presented in Fig. 4. The ranges for parameters γ and λ are set to [5, 10, 50, 100, 150] and [5, 10, 20, 50, 80, 100, 150, 200], respectively. We conduct a 5x8 grid search to explore the optimal parameter combinations. We observe

that OMELET is still sensitive to parameters, especially when $\gamma = [5, 10]$. Therefore, we recommend setting the parameter γ within the range [50, 100, 150] in practical scenarios. For the parameter λ , we observe that on the vast majority of datasets, selecting larger values yields superior performance. Only on certain datasets (e.g. ALLAML) does a smaller λ lead to better performance. Therefore, in practical scenarios, we recommend setting a larger λ ($\lambda \geq 50$).

Fig. 5 shows the impact of the base kernel parameter σ on the performance of OMELET. We observe that the parameter σ has less influence on the ensemble performance compared to parameters λ and γ . This indicates that the introduction of consensus kernel learning makes OMELET almost unaffected by specific kernel function parameters. Therefore, in real-world scenarios, the application of the OMELET method mainly depends on selecting appropriate parameters λ and γ .

Fig. 5. Performance of OMELET w.r.t. σ .Fig. 6. Performance of OMELET w.r.t. ensemble size m .

4.4. Ensemble size analysis

In practical applications, an excessive number of ensemble members can lead to increased computation time, and may affect ensemble performance. This is because ensemble performance is influenced by the diversity and accuracy of the base clustering result set [17]. Therefore, it is necessary to research the impact of an appropriate number of ensemble members on performance. Fig. 6 illustrates the comparative results of algorithm performance on different ensemble sizes. It is evident that OMELET outperforms the comparison methods at every ensemble size, reaffirming its superior performance. Especially on the **Ion.** dataset, when the ensemble size m of OMELET is 10, its performance surpasses that of all comparison methods at any ensemble size. While some ensemble clustering methods claim that ensemble performance improves with an increase in ensemble size, we observe that this improvement is not strictly monotonic. This phenomenon is evident even in many SOTA methods [14,52]. A possible reason is that increasing the number of base clustering results may impact the diversity and accuracy of the base clustering result set, thereby affecting the current performance. Therefore, our recommendation aligns with the common advice in most ensemble methods, suggesting a practical ensemble size m equals 20.

4.5. Ablation study

Table 7 shows the results of the ablation study on OMELET. The two key components of OMELET are multiple kernel learning and topological similarity learning. We naturally consider six degraded variants: re-

moving different types of base kernels in multi-kernel learning, including removing Gaussian kernels ('w/o Gaussian'), removing the linear kernel ('w/o Linear'), removing the polynomial kernel ('w/o Polynomial'), and removing adaptive weighting ('w/o $\alpha^{(i)}$ '); removing multiple kernel learning and applying topological similarity learning directly to the CA matrix ('w/o MK'); and removing topological similarity learning ('w/o Topo'). Overall, OMELET achieves the best performance on four datasets, validating the effectiveness of combining these components in a unified framework. Regarding the kernel components, removing any kernel type consistently degrades performance, indicating that each type contributes to the final results. Notably, removing the Gaussian kernels causes the largest drop, while removing the linear or polynomial kernel leads to relatively smaller degradations. Moreover, disabling the adaptive weighting parameter $\alpha^{(i)}$ also yields a clear performance decrease, confirming the benefit of adaptive kernel fusion. Regarding the denoising mechanism, removing topological similarity learning ('w/o Topo') results in a pronounced performance degradation. If multiple kernel learning is removed ('w/o MK'), performance still decreases, but the drop is less severe than that of 'w/o Topo'. This suggests that denoising directly in the CA matrix is challenging, whereas the multiple kernel (consensus kernel) space derived from CA matrix better separates noise from meaningful similarity relationships. In summary, although the CA matrix is inevitably noisy and multi-kernel mapping may further amplify this noise, the proposed topological-aware similarity learning mitigates the effect by leveraging the consensus relations learned from consensus-kernel learning and suppressing isolated spurious relations.

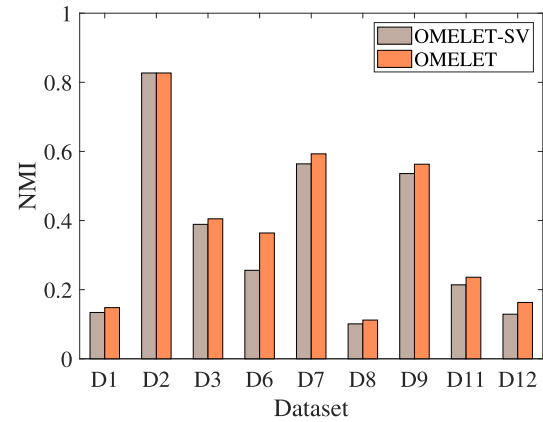
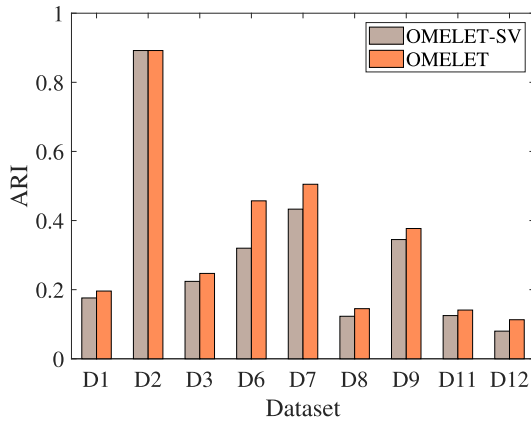
Table 7
Ablation study of the proposed OMELET (%).

Method	ARI					NMI				
	ALLAML	Mam.	Ion.	Agg.	Avg.	ALLAML	Mam.	Ion.	Agg.	Avg.
w/o Gaussian	18.0	11.9	34.7	98.9	40.9	14.4	9.7	27.3	98.5	37.5
w/o Linear	19.4	14.5	42.5	98.9	43.8	14.6	11.0	35.0	98.5	39.8
w/o Polynomial	19.3	13.7	43.5	98.9	43.9	14.5	10.9	35.3	98.5	39.8
w/o $\alpha^{(i)}$	18.8	8.1	16.3	98.9	35.5	14.7	7.5	12.7	98.4	33.3
w/o MK	15.4	10.5	20.8	99.0	36.4	13.3	9.8	18.1	98.5	34.9
w/o Topo	14.0	14.1	17.6	98.9	36.2	11.3	11.0	13.5	98.4	33.6
OMELET	19.6	14.5	45.7	99.0	44.7	14.8	11.0	36.4	98.5	40.2

Table 8

Running time of different clustering methods (in seconds).

Dataset	RSEC-Z	EPCMC	DREC	SCCBG	SPCE	CESHL	TRCE	CEAM	FSEC	SMKKM	OMELET-SV($p = 100$)	OMELET-SV($p = \lfloor 10\sqrt{n} \rfloor$)	OMELET
D2	12.476	0.027	0.116	80.765	2.160	2.553	1.307	16.757	0.011	4.244	0.123	0.635	2.722
D3	0.306	0.034	0.017	6.754	0.039	0.166	0.084	3.357	0.010	0.174	0.091	0.121	0.090
D4	16.279	0.041	0.066	79.213	2.753	2.652	2.664	22.193	0.020	7.280	0.128	0.711	3.908
D5	3.297	0.031	0.030	39.375	0.614	1.011	0.896	8.588	0.015	1.703	0.113	0.307	0.703
D6	2.346	0.018	0.033	32.540	0.295	0.688	0.507	6.357	0.032	0.579	0.103	0.305	0.581
D7	5.382	0.045	0.029	53.714	0.964	2.422	1.618	13.712	0.021	1.781	0.114	0.417	1.218
D8	20.273	0.032	0.035	87.170	3.163	3.730	3.196	21.091	0.013	7.214	0.125	0.731	4.698
D9	1.906	0.028	0.042	33.968	0.487	3.516	0.477	7.007	0.014	0.578	0.107	0.268	0.929
D10	1.618	0.022	0.040	23.387	0.234	0.469	0.336	5.405	0.010	0.442	0.083	0.285	0.458
D11	1482.401	0.271	3.732	1323.180	113.802	201.612	111.260	304.835	0.048	258.397	0.164	4.341	785.967
D12	195.661	0.123	0.078	516.941	24.012	30.427	16.448	80.298	0.027	31.481	0.110	1.904	78.235
D13	8667.582	1.071	201.765	23592.878	1766.980	711.333	749.630	2152.494	0.215	3856.769	0.602	21.535	6244.666



(a) ARI result

(b) NMI result

Fig. 7. Performance comparison between OMELET and OMELET-SV.

4.6. Effectiveness of the OMELET-SV

As shown in Fig. 7, we evaluate both the clustering accuracy of the scalable variant, OMELET-SV. On most datasets, OMELET-SV exhibits a slight drop in accuracy, which may be because it cannot fully capture the complete topological affinity graph among samples. Across these datasets, the average performance of all compared methods is 27.6% in ARI and 33.34% in NMI, whereas OMELET-SV improves these averages by 2.6% and 1.66%, respectively. The original OMELET achieves larger gains of 6.54% (ARI) and 4.56% (NMI). As shown in Table 8, we report the runtime of OMELET-SV. Compared with OMELET, OMELET-SV achieves a substantial improvement in efficiency. Although OMELET-SV is still less efficient than the highly optimized baseline FSEC, it remains competitive when considering the trade-off between accuracy and efficiency. Therefore, when higher accuracy is required in practice, OMELET is preferable. However, when considering runtime and trading

off accuracy for efficiency, OMELET-SV becomes the more advantageous choice.

5. Conclusion

In this paper, we propose OMELET, an ensemble clustering method based on multiple kernel and topological similarity learning. To address the issue of the single nature of sample similarity learning in existing methods, we introduce multiple kernel learning to capture sample similarity from multiple perspectives. We learn a consensus kernel from multiple base kernels, and parameter sensitivity experiments validate the insensitivity of the OMELET method to kernel function parameters. Then, we introduce topological similarity to remeasure the similarity relationships between samples, ultimately capturing the topological affinity matrix between the samples. Extensive experimental results demonstrate the effectiveness of OMELET. In addition, we introduce anchor points

and propose a scalable variant of the OMELET method, called OMELET-SV, which enables OMELET to be extended to large-scale datasets. However, like most unsupervised methods, OMELET's performance partially depends on hyperparameters. In the future, we will focus on reducing OMELET's sensitivity to hyperparameters to further improve its robustness.

CRedit authorship contribution statement

Jiaxuan Xu: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation; **Lei Duan:** Writing – review & editing, Resources, Methodology, Funding acquisition; **Xinye Wang:** Resources, Methodology, Data curation; **Liang Du:** Supervision, Methodology, Investigation.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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